SRCNN on KV260: experiment evolution：Phase 1b: streaming architecture and parameter sweep

Condensed from 231 registered experiments (results.csv). Only turning points are drawn; every ID links to experiments/<ID>_*/. Numbers are ms/frame; the evidence level is stated on each node.

Metric: ms/frame = cycles x clock period. HLS = HLS estimated cycles x target clock; RTL x routed = C/RTL co-simulation cycles x an independently closed post-route clock. No value is a board measurement.

Solid: baseline or parameter evolution, retry after fix, integration of an existing module; dashed: numerical evidence or inspiration, control, verification of the same design, replay. Edges are design lineage, not Git history.

passed, comparable metric (★ selected) | passed with warning | failed or below gate | invalid configuration | timeout or interrupted | planned, not run | analysis | partial experiment, no whole-design metric

E000 Original float baseline

E017 ap_fixed reference, 10-bit

E029 Directives (valid retry)

E033 10-bit streaming network, T=32

Line buffers and stream connections remove the full intermediate feature maps

HLS 123.5 ms

✗ LUT 136,868: narrow multiplications fell into LUTs, only 9 DSP used

E047 Force products onto DSP

One change: bind the 251 products per cycle to DSP48

HLS 29.1 ms

✗ 251 DSP as planned, but LUT 89,341 exceeds the cap

E039 Weights in LUTRAM

One change: weight storage from BRAM to LUTRAM

HLS 29.8 ms, 44,530 LUT

! First resource-feasible point; HLS timing warning

E043 Fastest point of the sweep

One axis per step: T, unroll split, weight storage, clock

HLS 6.94 ms

! HLS timing warning; phase-1 selected point

E070 First Vivado route

Route the first feasible point at the 10 ns target

routed: timing not met

✗ WNS −1.277 ns: an unregistered 8-deep DSP cascade. First calibration point: clock 1.5x worse than HLS, LUT 0.46x